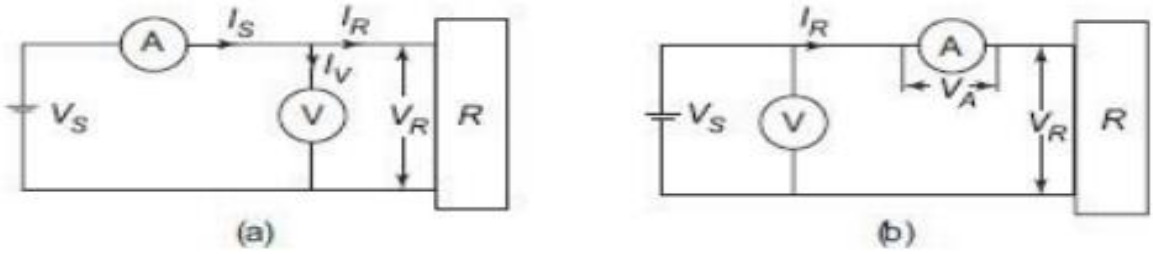
POWER MEASUREMENT

POWER MEASUREMENT IN dc CIRCUITS:

Electric power (P) consumed by a load (R) supplied from a dc power supply (V_S) is the product of the voltage across the load (V_R) and the current flowing through the load (I_R):

$$P = V_R \times I_R \tag{1}$$

Thus, power measurement in a dc circuit can be carried out using a voltmeter (V) and an ammeter (A) using any one of the arrangements shown in Figure

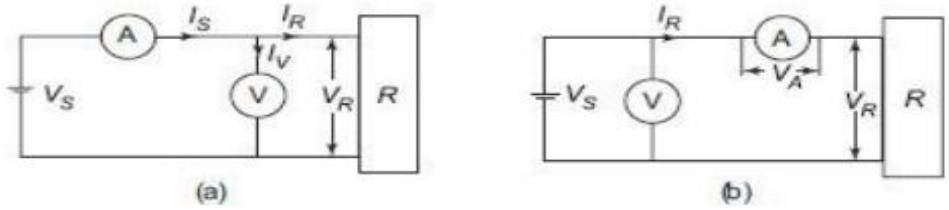


Both the voltmeter and the ammeter requires power for their own operations. In the arrangement of Figure (a), the voltmeter is connected between the load and the ammeter. The ammeter thus, in this case measures the current flowing into the voltmeter, in addition to the current flowing into the load

$$\text{Current through the voltmeter} = I_V = V_R/R_V$$
$$\text{where, } R_V \text{ is the internal resistance of the voltmeter.} \tag{2}$$

$$\begin{aligned}\text{Power consumed by the load} &= V_R \times I_R = V_R \times (I_S - I_V)\\ &= V_R \times I_S - V_R \times I_V\\ &= V_R \times I_S - V_R^2/R_V\\ &= \text{Power indicated by instruments} - \text{Power loss in voltmeter}\end{aligned} \tag{3}$$

Thus, Power indicated = Power consumed + Power loss in voltmeter



In the arrangement of Figure (b), the voltmeter measures the voltage drop across the ammeter in addition to that dropping across the load.

$$\text{Voltage drop across ammeter} = V_A = I_R \times R_A$$

where, R_A is the internal resistance of the ammeter. (4)

$$\begin{aligned}\text{Power consumed by the load} &= V_R \times I_R = (V_S - V_A) \times I_R \\ &= V_S \times I_R - V_A \times I_R \\ &= V_S \times I_R - I_R^2 \times R_A \\ &= \text{power indicated by instruments} - \text{Power loss in ammeter}\end{aligned} \quad (5)$$

Thus, Power indicated = Power consumed + Power loss in Ammeter.

Thus, both arrangements indicate the additional power absorbed by the instruments in addition to indicating the true power consumed by the load only. The corresponding measurement errors are generally referred to as insertion errors.

POWER MEASUREMENT IN ac CIRCUITS:

In alternating current circuits, the instantaneous power varies continuously as the voltage and current varies while going through a cycle. In such a case, the power at any instant is given by

$$p(t) = v(t) \times i(t) \quad (6)$$

where, p(t), v(t), and i(t) are values of instantaneous power, voltage, and current respectively. Thus, if both voltage and current can be assumed to be sinusoidal, with the current lagging the voltage by phase-angle φ, then

$$v(t) = V_m \sin \omega t$$

and $$i(t) = I_m \sin (\omega t - \varphi)$$

where, Vm and Im are peak values of voltage and current respectively, and w is the angular frequency. The instantaneous power p is therefore given by

$$p(t) = V_m I_m \sin \omega t \sin(\omega t - \varphi) \tag{7}$$

or, $$p(t) = \frac{V_m I_m}{2}[\cos\varphi - \cos(2\omega t - \varphi)]$$

Average value of power over a complete cycle in such a case will be

$$P = \frac{1}{2T}\int_0^{2T} p(t)\,dt = \frac{1}{2T}\int_0^{2T} \frac{V_m I_m}{2}[\cos\varphi - \cos(2\omega t - \varphi)]\,dt$$

$$= \frac{V_m I_m}{2T}\int_0^{2T}\left[\cos\varphi - \cos\left(\frac{4\pi}{T}t - \varphi\right)\right]dt$$

$$= \frac{V_m I_m}{2T}\left[\cos\varphi\, t\Big|_0^T - \frac{T}{4\pi}\sin\left(\frac{4\pi}{T}t - \varphi\right)\Big|_0^T\right]$$

$$= \frac{V_m I_m}{4T}[\cos\varphi T - 0]$$

$$= \frac{V_m I_m}{2}\cos\varphi$$

$$= \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}\cos\varphi$$

$$= V I \cos\varphi$$

cosφ (8)

where, Vand I are rms values of voltage and current respectively and is power factor of the load.

DYNAMOMETER-TYPE WATTMETER:

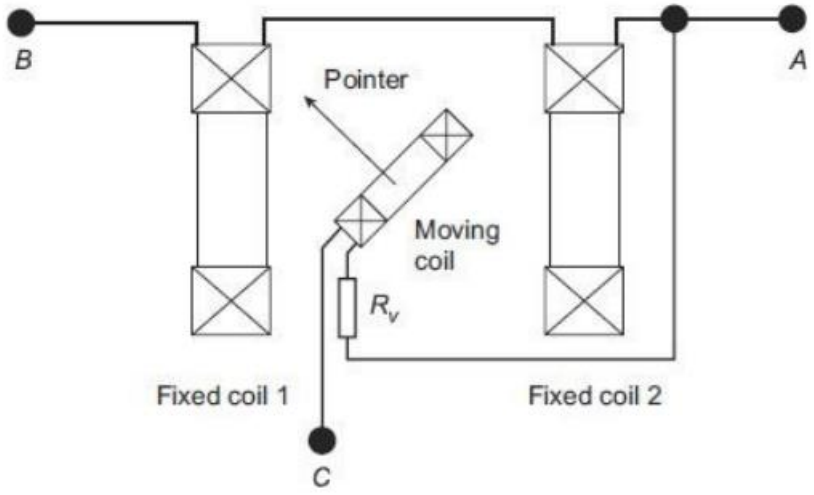
Power in dc circuits can also be measured by wattmeter. Wattmeter can give direct indication of power and there is no need to multiply two readings as in the case when ammeter and voltmeter is used.

The type of wattmeter most commonly used for such power measurement is the dynamometer. It is built by

(1) two fixed coils, connected in series and positioned coaxially with space between them, and

(2) a moving coil, placed between the fixed coils and fitted with a pointer.

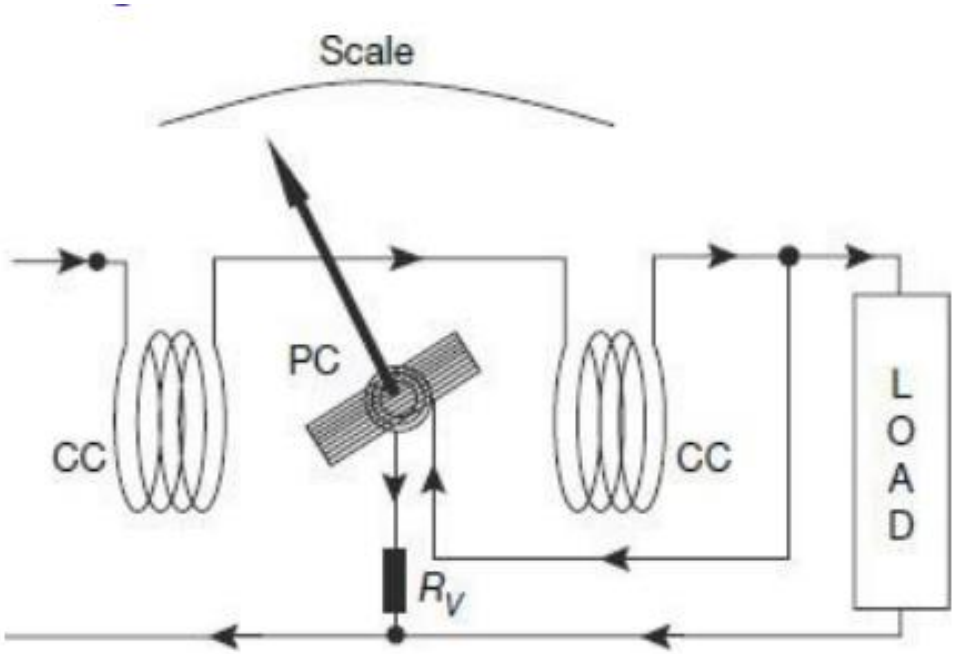
Such a construction for a dynamometer-type wattmeter is shown in Figure. It can be shown that the torque produced in the dynamometer is proportional to the product of the current flowing through the fixed coils times that through the moving coil. The fixed coils, generally referred to as current coils, carry the load current while the moving coil, generally referred to as voltage coil, carries a current that is proportional, via the multiplier resistor R_V , to the voltage across the load resistor R. As a consequence, the deflection of the moving coil is proportional to the power consumed by the load.



CONSTRUCTION:

1. Fixed colis
2. Moving coils
3. Control: spring control
4. Damping: air friction
5. Scales and pointers: are equipped with mirror type scales and knife edge pointers to remove reading errors due to parallax.

OPERATION:



V = voltage to be measured (rms)

I = current to be measured (rms)

i_P = voltage (pressure) coil instantaneous current

i_C = current coil instantaneous current

R_V = external resistance connected with pressure coil

R_P = resistance of pressure coil circuit (PC resistance + R_V)

M = mutual inductance between current coil and pressure coil

θ = angle of deflection of the moving system

ω = angular frequency of supply in radians per second

φ = phase-angle lag of current I with respect to voltage V

the instantaneous torque of the electrodynamometer wattmeter shown in Figure is given by

$$T_i = i_P i_C \frac{dM}{d\theta} \tag{9}$$

Instantaneous value of voltage across the pressure-coil circuit is

$$v_P = \sqrt{2} \times V \sin \omega t$$

If the pressure coil resistance can be assumed to be very high, the whole pressure coil can be assumed to be behaving like a resistance only. The current i_p in the pressure coil thus, can be assumed to in phase with the voltage v_p, and its instantaneous value is

$$i_p = {}^{v_p}\!/_{R_p} = \sqrt{2} \times {}^{V}\!/_{R_p} \sin\omega t = \sqrt{2} \times I_p \sin\omega t$$

where $I_p = V/R_p$ is the rms value of current in pressure coil.

Assuming that the pressure-coil resistance is sufficiently high to prevent branching out of any portion of the supply current towards the pressure coil, the current coil current can be written as

$$i_C = \sqrt{2} \times I \sin(\omega t - \varphi)$$

Thus, instantaneous torque from (9) can be written as

$$\begin{aligned} T_i &= \sqrt{2} \times I_p \sin\omega t \times \sqrt{2} \times I \sin(\omega t - \varphi) \frac{dM}{d\theta} \\ &= 2 I_p I \sin\omega t \sin(\omega t - \varphi) \frac{dM}{d\theta} \\ &= I_p I \{\cos\varphi - \cos(2\omega t - \varphi)\} \frac{dM}{d\theta} \end{aligned} \tag{10}$$

Presence of the term containing 2wt, indicates the instantaneous torque as shown in (10) varies at twice the frequency of voltage and current. Average deflecting torque over a complete cycle is

$$\begin{aligned} T_d &= \frac{1}{T}\int_0^T T_i \, d\omega t = \frac{1}{2\pi}\int_0^{2\pi} I_p I \{\cos\varphi - \cos(2\omega t - \varphi)\} \frac{dM}{d\theta} \, d\omega t \\ &= \frac{I_p I}{2\pi} \left[\omega t \cos\varphi\right]_0^{2\pi} \frac{dM}{d\theta} \\ &= I_p I \cos\varphi \frac{dM}{d\theta} \end{aligned} \tag{11}$$

$$\begin{aligned} &= \frac{V}{R_P} I \cos\varphi \frac{dM}{d\theta} \\ &= \frac{VI\cos\varphi}{R_P} \frac{dM}{d\theta} \end{aligned} \tag{12}$$

With a spring constant K, the controlling torque provided by the spring for a final steady-state deflection of θ is given by $T_C = K\theta$

Under steady-state condition, the average deflecting torque will be balanced by the controlling torque provided by the spring. Thus, at balanced condition $T_C = Td$

$$T_C = T_d$$

$$K\theta = \frac{VI\cos\varphi}{R_P}\frac{dM}{d\theta}$$

$$\theta = \frac{VI\cos\varphi}{KR_P}\frac{dM}{d\theta}$$

$$\theta = \left(K_1\frac{dM}{d\theta}\right)P \qquad (13)$$

where, P is the power to be measured and $K1 = 1/KR_P$ is a constant. Steady-state deflection θ is thus found to be an indication of the power P to be measured.

ERRORS IN ELECTRODYNAMOMETER-TYPE WATTMETER:

1. Error due to Pressure-Coil Inductance:

To have an estimate of such error, let us consider the following:

V = voltage applied to the pressure coil circuit (rms)

I = current in the current coil circuit (rms)

I_P = current in the voltage (pressure) coil circuit (rms)

r_P = resistance of pressure coil only

L = inductance of pressure coil

R_V = external resistance connected with pressure coil

R_P = resistance of pressure coil circuit (PC resistance + R_V)

Z_P = impedance of pressure coil circuit

M = mutual inductance between current coil and pressure coil

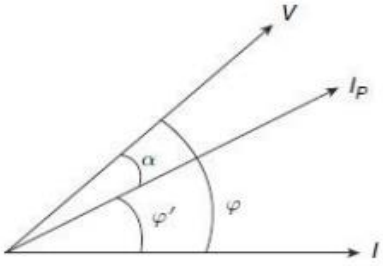
ω = angular frequency of supply in radian per second

φ = phase-angle lag of current I with respect to voltage V

Due to inherent inductance of the pressure coil circuit, the current and voltage in the pressure coil will no longer be in phase, rather the current through the pressure coil will lag the voltage across it by a certain angle given by

$$\alpha = \tan^{-1}\left(\frac{\omega L}{R_P}\right) = \tan^{-1}\left(\frac{\omega L}{r_P + R_V}\right)$$

As can be seen from Figure below, current through the pressure coil lags voltage across it by a phase-angle which is less than that between the current coil current and the pressure coil voltage.



Wattmeter phasor diagram with pressure coil inductance

In such a case, phase-angle difference between the with pressure coil inductance, pressure coil current and current coil current is

$$\varphi' = \varphi - \alpha$$

$$\theta' = \frac{I_P I}{K} \cos\varphi' \cdot \frac{dM}{d\theta}$$

$$\theta' = \frac{V}{Z_P K} I \cos(\varphi - \alpha) \cdot \frac{dM}{d\theta} \qquad (14)$$

Following from (11), the wattmeter deflection will be Relating to $R_P = Z_P \cos\alpha$ in the pressure coil circuit, the wattmeter deflection can be rewritten as VI dM

$$\theta' = \frac{VI}{R_P K} \cos\alpha \cdot \cos(\varphi - \alpha) \cdot \frac{dM}{d\theta}$$

In the absence of inductance, $Z_P = R_P$ and $\alpha = 0$; wattmeter in that case will read true power, given by,

$$\theta = \frac{VI}{R_P K} \cos\varphi \frac{dM}{d\theta} \qquad (15)$$

Taking the ratio of true power indication to actual wattmeter reading, we get

$$\frac{\text{True power indication}}{\text{Actual wattmeter reading}} = \frac{\theta}{\theta'} = \frac{\frac{VI}{R_P K} \cos\varphi \frac{dM}{d\theta}}{\frac{VI}{R_P K} \cos\alpha \cdot \cos(\varphi - \alpha) \cdot \frac{dM}{d\theta}} = \frac{\cos\varphi}{\cos\alpha \cdot \cos(\varphi - \alpha)}$$

Thus, the correction factor can be identified as

$$CF = \frac{\cos\varphi}{\cos\alpha \cdot \cos(\varphi - \alpha)}$$

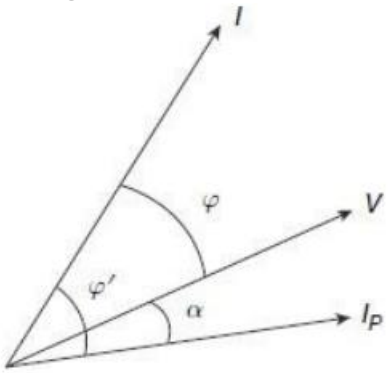
True power indication can thus be obtained from the actual wattmeter reading using the correction factor CF as True power indication = CF X Actual wattmeter reading

Thus, for lagging power factor loads,

$$\text{True power indication} = \frac{\cos\varphi}{\cos\alpha \cdot \cos(\varphi - \alpha)} \times \text{Actual wattmeter reading}_S$$

The above relations, along with Figure indicate that under lagging power factor loads, unless special precautions are taken, actual wattmeter reading will tend to display higher values as compared to true power consumed.

For leading power factor loads, however, the wattmeter phasor diagram will be as shown in below figure



Wattmeter phasor diagram with pressure coil inductance during leading load

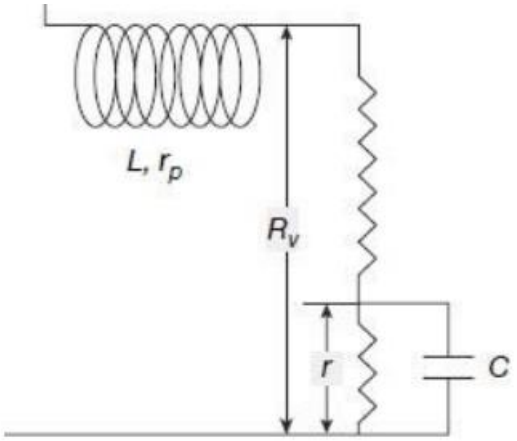
For leading power factor loads,

$$\text{True power indication} = \frac{\cos\varphi}{\cos\alpha \cdot \cos(\varphi + \alpha)} \times \text{Actual wattmeter reading}_S$$

The above relations, along with Figure (lagging pf) indicate that under leading power factor loads, unless special precautions are taken, actual wattmeter reading will tend to display higher values as compared to true power consumed.

2. Compensation for pressure coil inductance:

A wattmeter can be compensated for pressure coil inductance by connecting a preset value of capacitance across a certain portion of the external resistance connected in series with the pressure coil, as shown in Figure



Compensation for pressure coil inductance

The total impedance of the circuit in such a case can be written as

$$Z_p = (r_p + R_V - r) + j\omega L + \frac{r - j\omega Cr^2}{1 + \omega^2 C^2 r^2}$$

To make the entire circuit behave as purely resistive, if we can design the circuit parameters in such a case that for power frequencies

$$w^2C^2r^2 << 1$$

Then we can re-write the total impedance of the pressure coil as

$$Z_P = (r_p + R_V - r) + jwL + r - jwCr^2 = r_P + R_V + jw(L - Cr^2)$$

If by proper design, we can make $L = Cr^2$

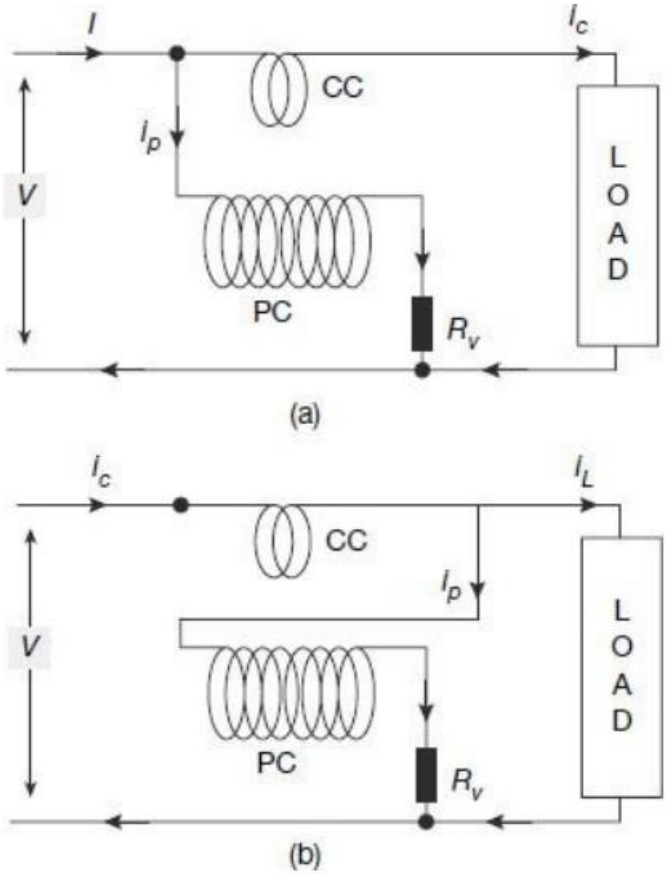
Then, impedance = $r_p + R_V = R_P$

Thus error introduced due pressure coil inductance can be substantially eliminated.

3. Error due to Pressure Coil Capacitance

The voltage, or pressure coil circuit may have inherent capacitance in addition to inductance. This capacitance effect is mainly due to inter-turn capacitance of the winding and external series resistance. The effect of stray capacitance of the pressure coil is opposite to that due to inductance. Therefore, the wattmeter reads low on lagging power factors and high on leading power factors of the load. Actual reading of the wattmeter, thus, once again needs to be corrected by the corresponding correction factors to obtain the true reading. The effect of capacitance (as well as inductance) varies with variable frequency of the supply.

4. Error due to Connection: There are two alternate methods of connection of wattmeter to the circuit for measurement of power. These are shown in Figure. In either of these connection modes, errors are introduced in measurement due power losses in pressure coil and current coil



Wattmeter connections

In the connection of Figure (a), the pressure coil is connected across the supply, thus pressure coil measures the voltage across the load, plus the voltage drop across the current coil. Wattmeter reading in this case will thus include power loss in current coil as well, along with power consumed by the load.

Wattmeter reading = Power consumed by load + Power loss in CC

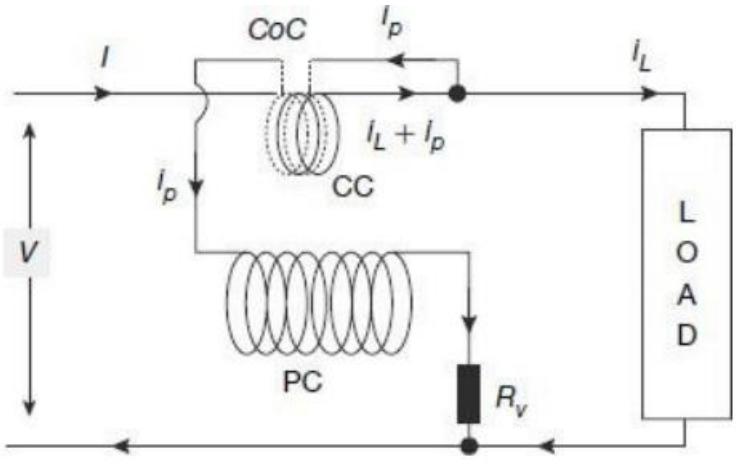
In the connection of Figure(b), the current coil is connected to the supply side; therefore, it carries load current plus the pressure coil current. Hence, wattmeter reading in this case includes, along with power consumed by the load, power loss in the pressure coil as well.

Wattmeter reading = Power consumed by load + Power loss in PC

In the case when load current is small, power loss in the current coil is small and hence the connection of Figure (a) will introduce comparatively less error in measurement.

On the other hand, when load current is large, current branching through the pressure coil is relatively small and error in measurement will be less if connection of Figure(b) is used.

Errors due to branching out of current through the pressure coil can be minimised by the use of compensating coil as schematically shown in Figure below



Schematic connection diagram of compensated wattmeter

In the compensated connection, the current coil consists of two windings, each winding having the same number of turns. The two windings are made as far as possible identical and coincident. One of the two windings (CC) is made of heavy wire that carries the load current plus the current for the pressure coil. The other winding (compensating coil–CoC) which is connected in series with the pressure coil, uses thin wire and carries only the current to the pressure coil. This current in the compensating coil is, however, in a direction opposite to the current in main current coil, creating a flux that opposes the main flux. The resultant magnetic field is thus due to the current coil only, effects of pressure coil current on the current coil flux mutually nullifying each other. Thus, error due to pressure coil current flowing in the current coil in cancelled out and the wattmeter indicates correct power.

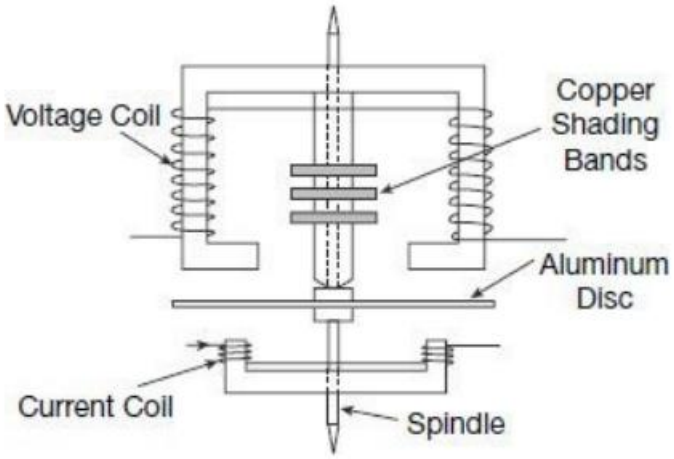
5.Error Caused by Vibration of the Moving System:

The instantaneous torque on the moving system varies cyclically at twice the frequency of the voltage and current (7.10). If any part of the moving system, such as the spring or the pointer has natural frequency close to that of torque pulsation, then accidental resonance may take place. In such a case, the moving system may vibrate with considerable amplitude. These vibrations may pose problems while noting the pointer position on the scale. These errors due to vibrations may be avoided by designing the moving elements to have natural frequencies much further away from twice the frequency of the supply voltage.

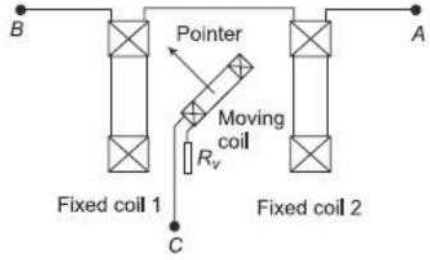
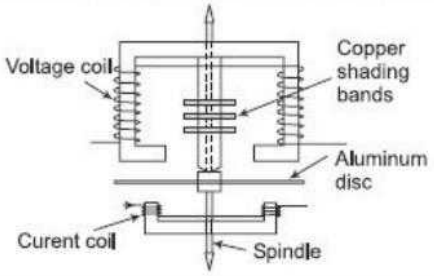
6. Temperature Errors:

Temperature changes may affect accuracy of wattmeter by altering the coil resistances. Temperature my change due to change in room temperature or even due to heating effects in conductors with flow of current. Change in temperature also affects the spring stiffness, thereby introducing error in the deflection process. High-precision instruments are fitted with temperature compensating resistors that tend to neutralise the effects of temperature variation.

INDUCTION-TYPE WATTMETER:

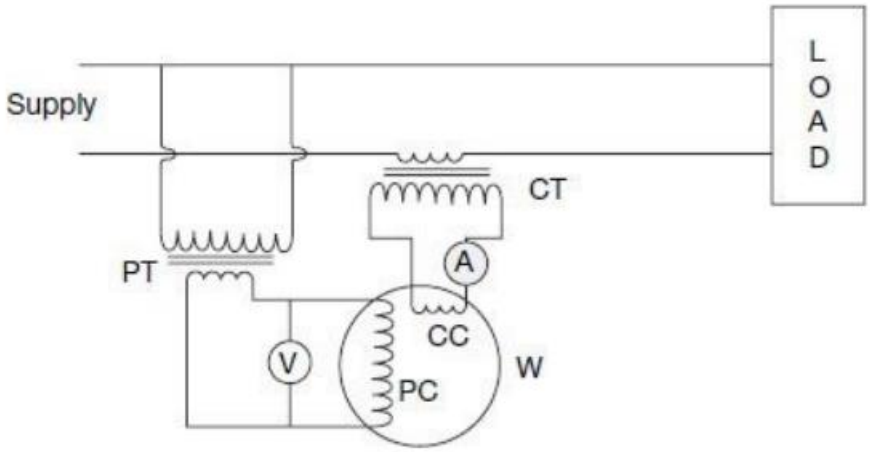


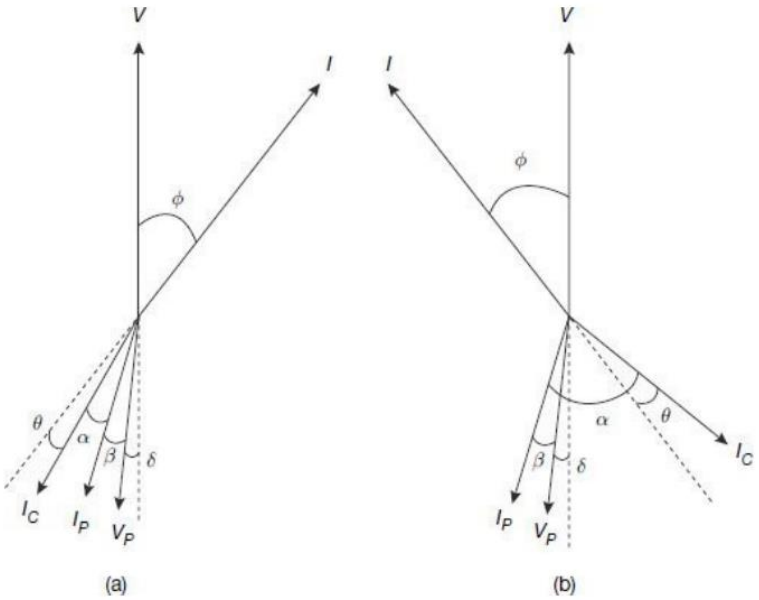
Induction-type wattmeters have two laminated iron-core electromagnets. One of the electromagnets is excited by the load current, and the other by a current proportional to the voltage of the circuit in which the power is to be measured. The upper magnet in Figure 7.19, which is connected across the voltage to be measured, is named as the shunt magnet, whereas the other electromagnet connected in series with the load to carry load current is called the series magnet. A thin aluminum disc, mounted in the space between the two magnets is acted upon by a combined effect of fluxes coming out of these two electromagnets. In ac circuits, interaction of these changing fluxes will induce Eddycurrent within the aluminum disc. The two voltage coils, connected in series, are wound in such a way that both of them send flux through the central limb. Copper shading bands fitted on the central limb of the shunt magnet makes the flux coming out of the magnet lag behind the applied voltage by 90°. The series magnet houses two small current coils in series. These are wound in a way that the fluxes they create within the core of the magnet are in the same direction.

Dynamometer Type Watteter	Induction Type Wattmeter
	
Schematic of electrodynamometer type wattmeter	Schematic of induction type wattmeter
Current coil split in two parts, but a single pressure coil	Both current and pressure coils split in two parts each, placed on each of two arms of the two magnets
Pressure coil is the moving coil	None of the coils are moving, rather there is an aluminum disc placed between the two electromagnets, that moves
Pointer is attached with the moving (pressure) coil	Pointer is attached with the aluminum disc
Can be used for measurement of power both in AC as well as DC circuits	Can only be used for measurement of power in AC circuits
Fluid friction damping is used	Eddy current damping is used
Both the coils are air-cored	Both the coils are mounted on laminated iron core
Can be used in circuits even with fluctuating frequency and voltage	Can be used only with circuits having relatively steady values of frequency and voltage

POWER MEASUREMENT USING INSTRUMENT TRANSFORMER

Power measurements in high-voltage, high-current circuits can be effectively done by the use of Potential Transformers (PT) and Current Transformers (CT) in conjunction with conventional wattmeters. By using a number of current transformers of different ratio for the current coil, and a number of potential transformers with different turns ratio for the voltage coil, the same wattmeter can be utilised for power measurements over a wide range. Primary winding of the CT is connected in series with the load and the secondary winding is connected with the wattmeter current coil, often with an ammeter in between. Primary winding of the PT is connected across the main power supply lines, and its secondary winding is connected in parallel with the wattmeter voltage coil. The connections of a wattmeter when so used are shown in Figure , an ammeter and a voltmeter is also connected in the circuit.





1. Lagging Power Factor

In general, phase-angle (θ) of CT is positive, whereas phase-angle of PT (δ) may be positive or negative.

Thus, phase-angle of the load is $\Phi = \theta + \alpha + \beta \pm \delta$

2. Leading Power Factor

Phase-angle of the load is $\Phi = \alpha - \theta - \beta \pm \delta$

3. Correction Factors

For the time being, neglecting the ratio error, the correction factor that need to be incorporated for lagging power factor load is

$$K = \frac{\cos\phi}{\cos\beta \cdot \cos\alpha} = \frac{\cos\phi}{\cos\beta \cdot \cos(\phi - \theta - \beta \pm \delta)}$$

For leading power factor, the correction factor is

$$K = \frac{\cos\phi}{\cos\beta \cdot \cos\alpha} = \frac{\cos\phi}{\cos\beta \cdot \cos(\phi + \theta + \beta \pm \delta)}$$

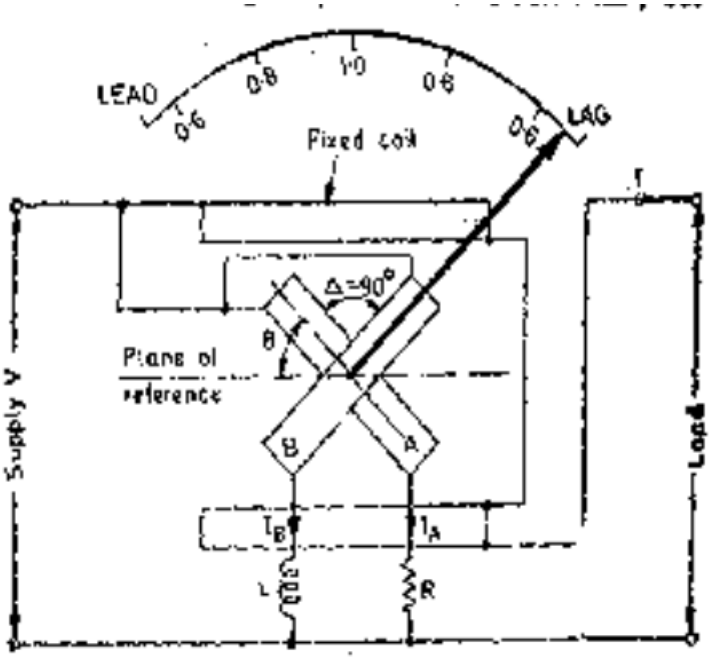
Considering ratio errors for both PT and CT, the corresponding correction factors for ratio errors also need to be incorporated.

General expression for the power to be measured can now be written as

Power = K x Actual PT ratio x Actual CT ratio x Wattmeter reading

POWER FACTOR METER

Single Phase Electrodynamometer Power Factor Meter



LOW PF METER:

The instrument that measures the low value of power factor accurately is known as the **Low Power Factor Wattmeter (LPFW).** The low power factor meter is used for measuring the power of the highly inductive circuit. It is also used in the resistive circuit which has a power factor range from 0.5 to 1.

What is the need of LPFW?

The ordinary Wattmeter used for measuring the low power factor gives the inaccurate result. This happens because of two reasons.

1. In low power factor meter, the magnitude of deflecting torque on moving coil is small even after the full excitation of the pressure and current coil.
2. The error occurs in the reading because of the pressure coil inductance.

Some additional features are added on the ordinary Wattmeter so that the meter can measure the power of the low power factor circuit.

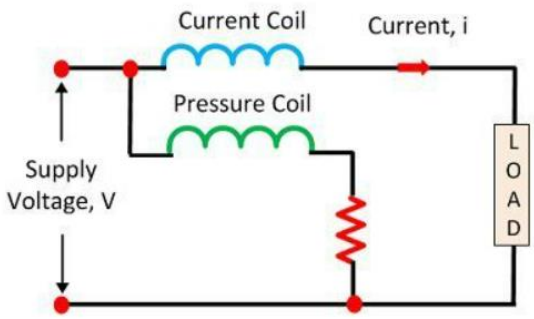
Modifications in Ordinary Wattmeter

Pressure Coil Current

The **pressure coil is designed for having a low value of resistance** so that the high value of current passes through it. This current produces the deflecting torque on the moving coil.

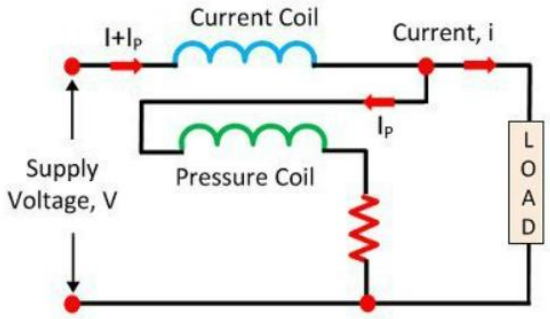
Compensation for Pressure Coil

There are two substitute methods of connecting the Wattmeter in the circuit. The figure below shows these two methods.



(a). Wattmeter Connection

In figure (a) the pressure coil is parallel connected to the supply voltage. And the current coil is connected in series with the supply voltage. The magnitude of the voltage across the pressure coil is equal to the supplied voltage. **The total power measured by the pressure coil is equal to the sum of the power loss in the load and the power loss in the current coil.**

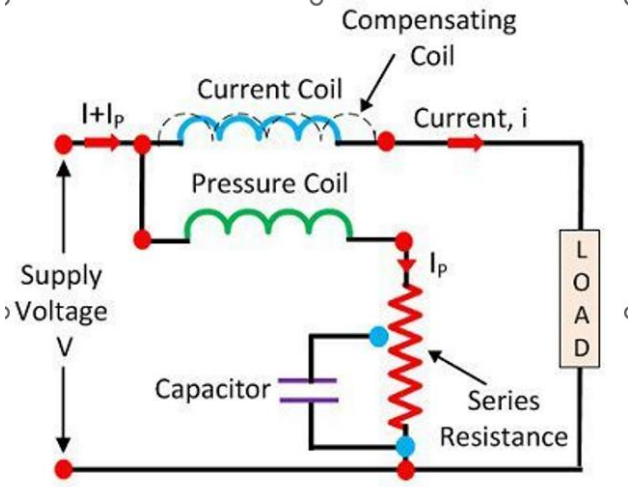


(b) Wattmeter Connections

In figure (b) the pressure coil is not connected in parallel with the load. Thereby, the magnitude of the pressure coil voltage is not equal to the supplied voltage. The output power obtains from the circuit is equal to the sum of the load power loss and the power loss of the pressure coil.

In low power circuit, the value of current is high, and that of the power is low. The high-value current causes the error in the Wattmeter reading. For reducing the error, the compensation coil is used in the circuit. The compensating coil

compensates the error in the circuit which induces because of low power factor.



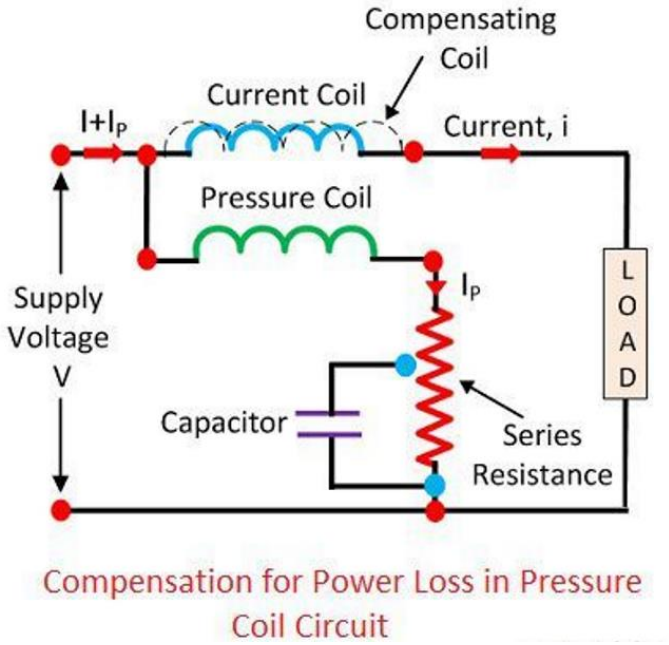
Compensation for Power Loss in Pressure Coil Circuit

Compensation For Inductance of Pressure Coil

The small amount of inductances is present in the pressure coil of the Wattmeter. This inductance causes the error in the reading. The error occurs in the pressure coil is given by the expression

$VISin\varphi tan\beta$

The φ is the angle between the pressure and the current coil. For a small value power factor φ is large. The large value of φ causes a large error in the reading. The compensating coil is used in the circuit for compensating the inductance error occurs in the Wattmeter. Along with the compensating coil, the capacitor is used in the circuit. The capacitor is placed in parallel with the pressure coil resistance.



Compensation for Power Loss in Pressure Coil Circuit

The capacitor eliminates the error. The compensation caused by the capacitor is slightly affected by the variation in frequencies. Above 10kHz the compensation is accurately held in the Wattmeter.

Small Control Torque

The control torque of the low power factor Wattmeter is kept small so that the full-scale deflection is obtained even for the small value of the power factor.

DIFF BETWEEN LPF AND UPF:

LPF wattmeter is used to measure power in inductive circuits. Inductive circuits have the property of lagging power factor and hence these wattmeters are used. Resistive circuits have unity power factor hence UPF wattmeters are used to measure power in these circuits. In open circuit test the secondary winding of transformer is kept open.As the secondary side is open, the entire coil will be purely inductive in nature.So the power will be lagging due to inductive property of the circuit. So LPF(Low Power Factor) Wattmeter is used in OPEN CIRCUIT TEST of Transformer. In short circuit test the secondary winding of transformer is short circuited. As secondary side is short circuited the entire coil will be purely resistive in nature. So the power factor

will be unity. So UPF(Unit Power Factor) Wattmeter is used in SHORT CIRCUIT TEST of Transformer.